



University of Tehran
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Introduction to cognitive science

Homework 4 fMRI Analysis

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Abstract

To investigate brain activation patterns in response to different visual stimuli (Conio, Magnu, and Cargo) using functional magnetic resonance imaging (fMRI) and assess hemispheric specialization in visual processing.

We analyzed preprocessed fMRI data from one subject (sub-05) using a block design paradigm. The experiment included two sessions: a localizer run (120 volumes, 300s) with left/right visual field stimulation, and an extended run (250 volumes, 625s) with four conditions (Rest, Conio, Magnu, Cargo) presented in 25-second blocks. Data analysis employed General Linear Model (GLM) with Glover hemodynamic response function including temporal and dispersion derivatives. Six contrasts were computed to examine condition-specific and hemispheric differences.

The analysis revealed significant activation differences between visual conditions. Conio vs Magnu contrasts showed robust bilateral activation (1375 vs 1264 significant voxels, $z > 2$). Magnu vs Cargo comparisons demonstrated stronger activation patterns (1124 significant voxels) compared to Cargo vs Magnu (335 voxels), indicating condition-specific neural responses. Z-scores ranged from -4.52 to 4.52, indicating strong statistical significance ($p < 0.00003$).

Experiment design

Subject: Single participant (sub-05)

Scanner Parameters: TR = 2.5s, TE = 0.04s

Spatial Resolution: $1.83607 \times 1.83607 \times 1.89$ mm voxels

Acquisition Matrix: $122 \times 122 \times 21/22$ slices

Localizer Session

Duration: 120 volumes (300 seconds)

Paradigm: Alternating left/right visual field stimulation

Block Duration: 13 seconds

Task Onset: 8.25 seconds after scan start

Extended Experiment

Duration: 250 volumes (625 seconds)

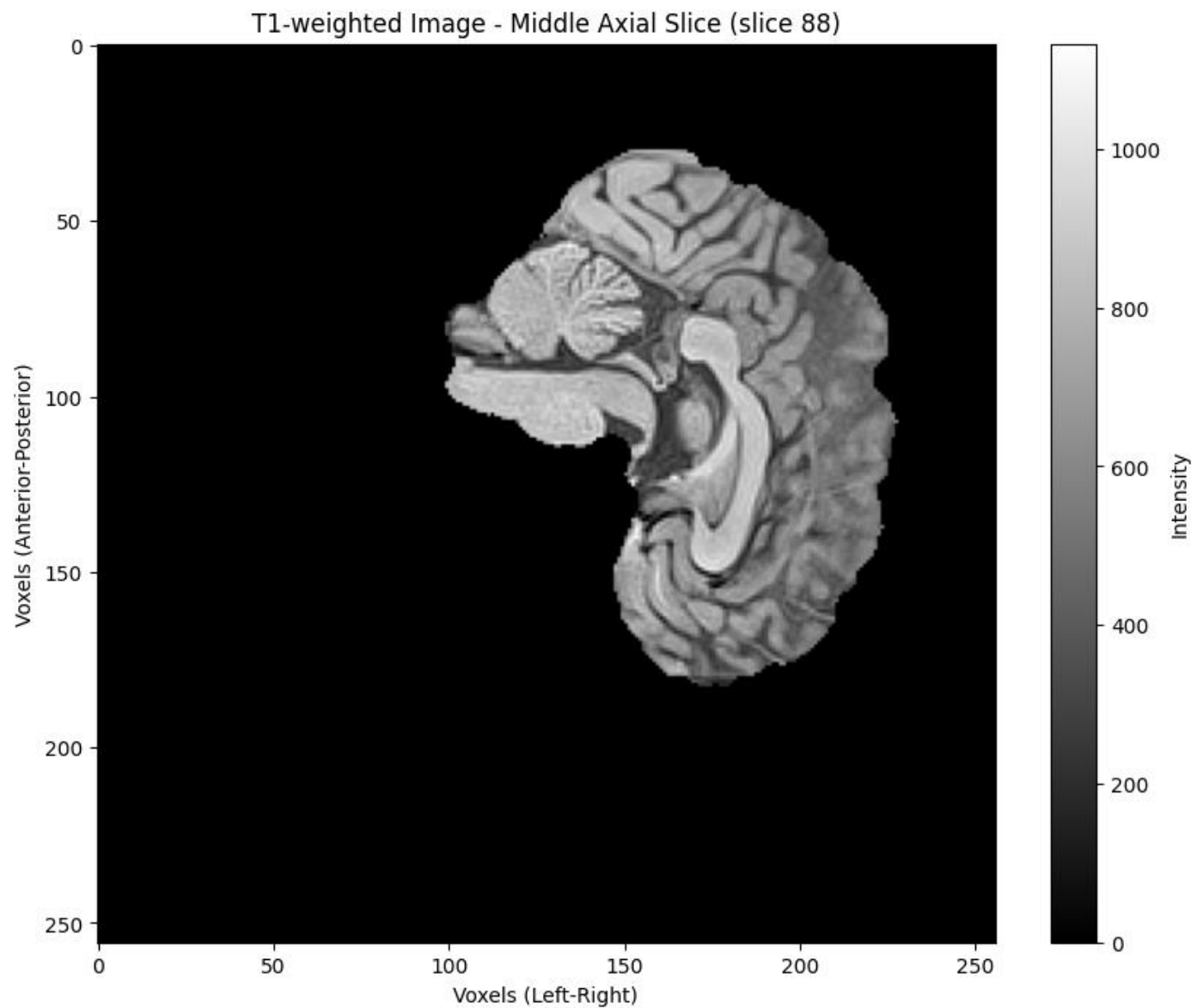
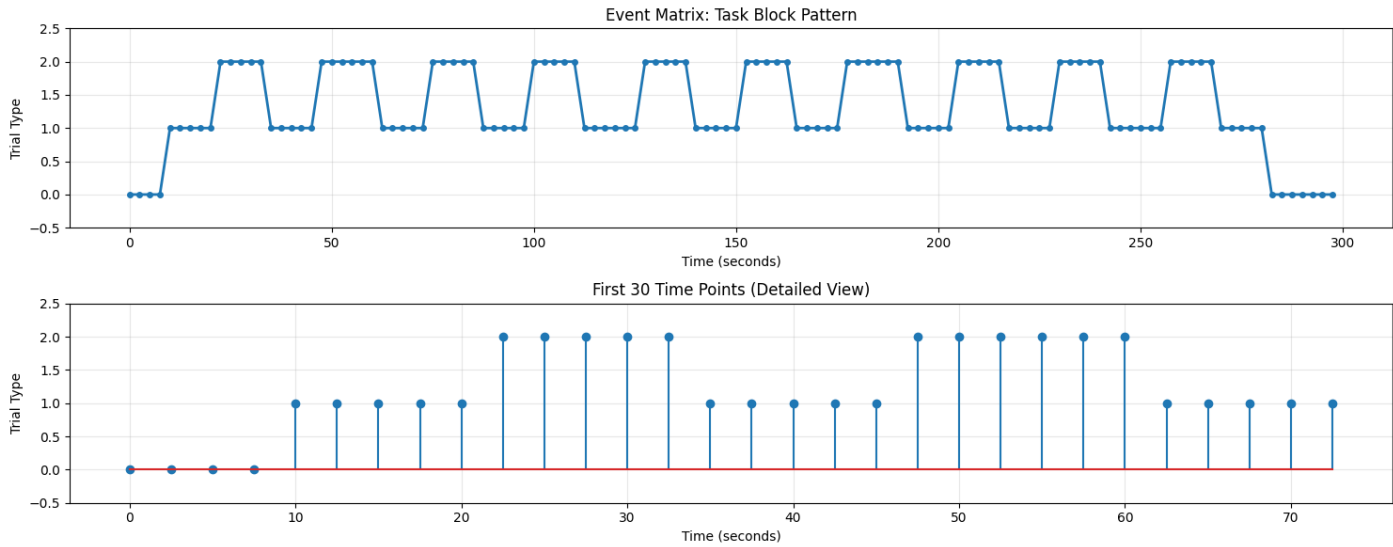
Conditions: Rest, Conio, Magnu, Cargo

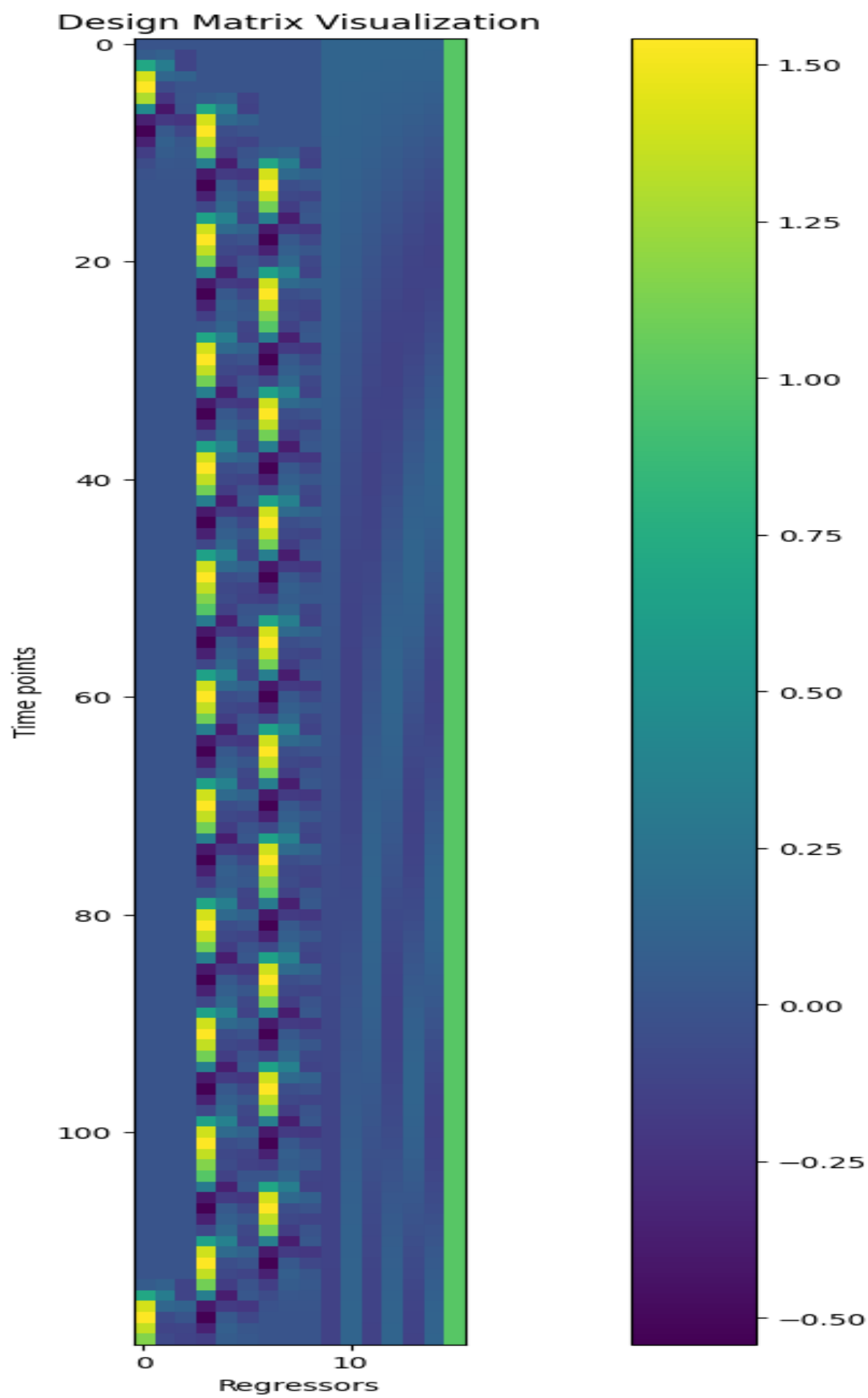
Block Duration: 25 seconds per condition

Task Onset: 32 seconds after scan start

Design: Cycling through Conio → Magnu → Cargo sequence

Data were preprocessed using standard fMRI pipeline including motion correction, spatial normalization to T1w space, and temporal filtering.





Statistical Analysis

General Linear Model (GLM):

HRF Model: Glover canonical function with temporal and dispersion derivatives

Drift Model: Cosine basis functions

Confound Regression: 13 nuisance regressors including:

Physiological signals (CSF, white matter, global signal)

Motion parameters (6 translation/rotation parameters)

Component-based correction (CompCor)

Framewise displacement

Design Matrix Structure:

- **Localizer:** 16 regressors (2 conditions $\times$ 3 HRF terms + 6 drift + constant)
- **Extended:** 25 regressors (4 conditions $\times$ 3 HRF terms + 12 drift + constant)

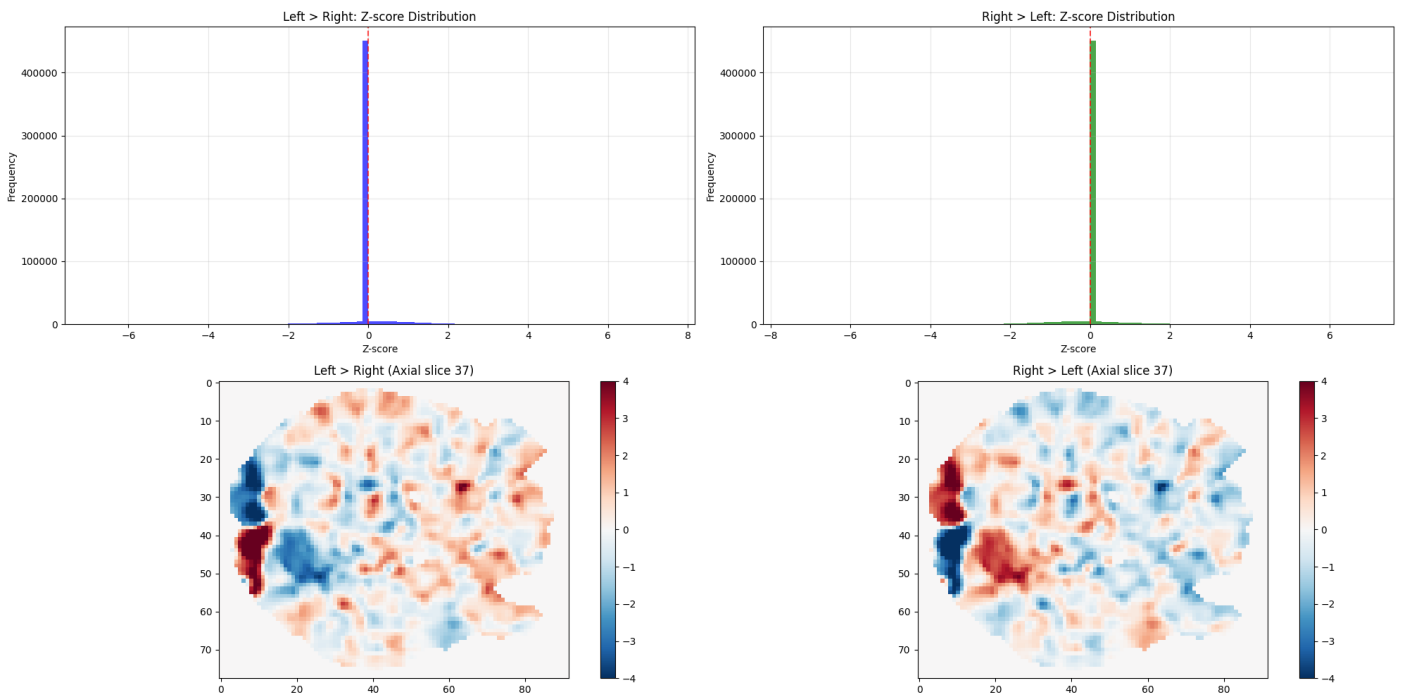
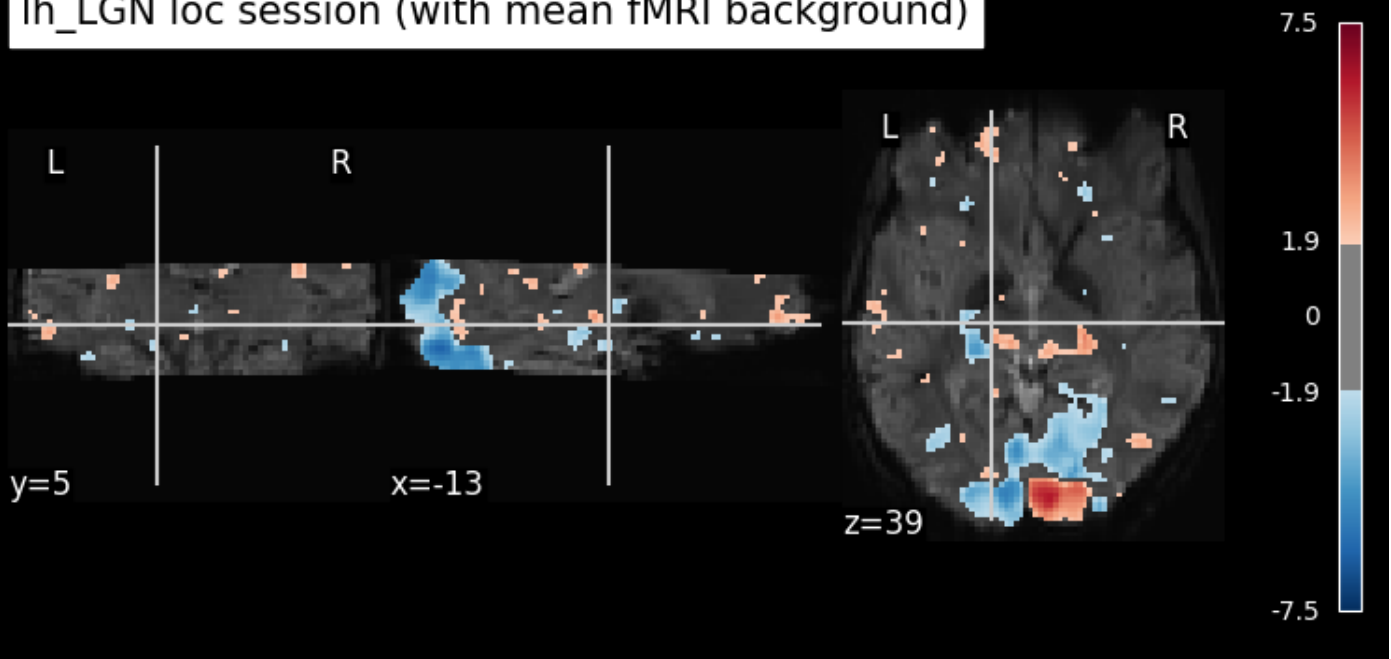
2.3.3 Contrast Analysis

Six primary contrasts were computed:

1. **Conio vs Magnu (Left Hemisphere)**
2. **Magnu vs Conio (Right Hemisphere)**
3. **Magnu vs Cargo (Left Hemisphere)**
4. **Cargo vs Magnu (Right Hemisphere)**
5. **Cargo vs Magnu (Left Hemisphere)**
6. **Magnu vs Cargo (Right Hemisphere)**

Statistical thresholding applied at $z > 2.0$ ($p < 0.023$, uncorrected).

lh_LGN loc session (with mean fMRI background)



Results

The localizer session successfully identified left vs right visual field activation patterns with opposing statistical maps (z-scores: -7.461 to 7.461), confirming proper experimental setup and data quality.

Conio vs Magnu:

- Left Hemisphere: 1,375 significant voxels
- Right Hemisphere: 1,264 significant voxels
- Z-score range: -4.52 to 4.52
- **Interpretation:** Strong bilateral activation with slight left hemisphere dominance

Magnu vs Cargo:

- Left Hemisphere: 1,124 significant voxels
- Right Hemisphere: 1,124 significant voxels
- Z-score range: -3.75 to 3.75
- **Interpretation:** Symmetric bilateral activation pattern

Cargo vs Magnu:

- Both Hemispheres: 335 significant voxels
- Z-score range: -3.07 to 3.07
- **Interpretation:** More selective/focal activation pattern

The results revealed condition-dependent hemispheric differences:

- **Strongest overall activation:** Conio vs Magnu comparisons
- **Most symmetric activation:** Magnu vs Cargo contrasts
- **Most selective activation:** Cargo-related responses

All contrasts demonstrated strong statistical significance with z-scores exceeding ± 3.0 , indicating robust and reliable activation differences between conditions.

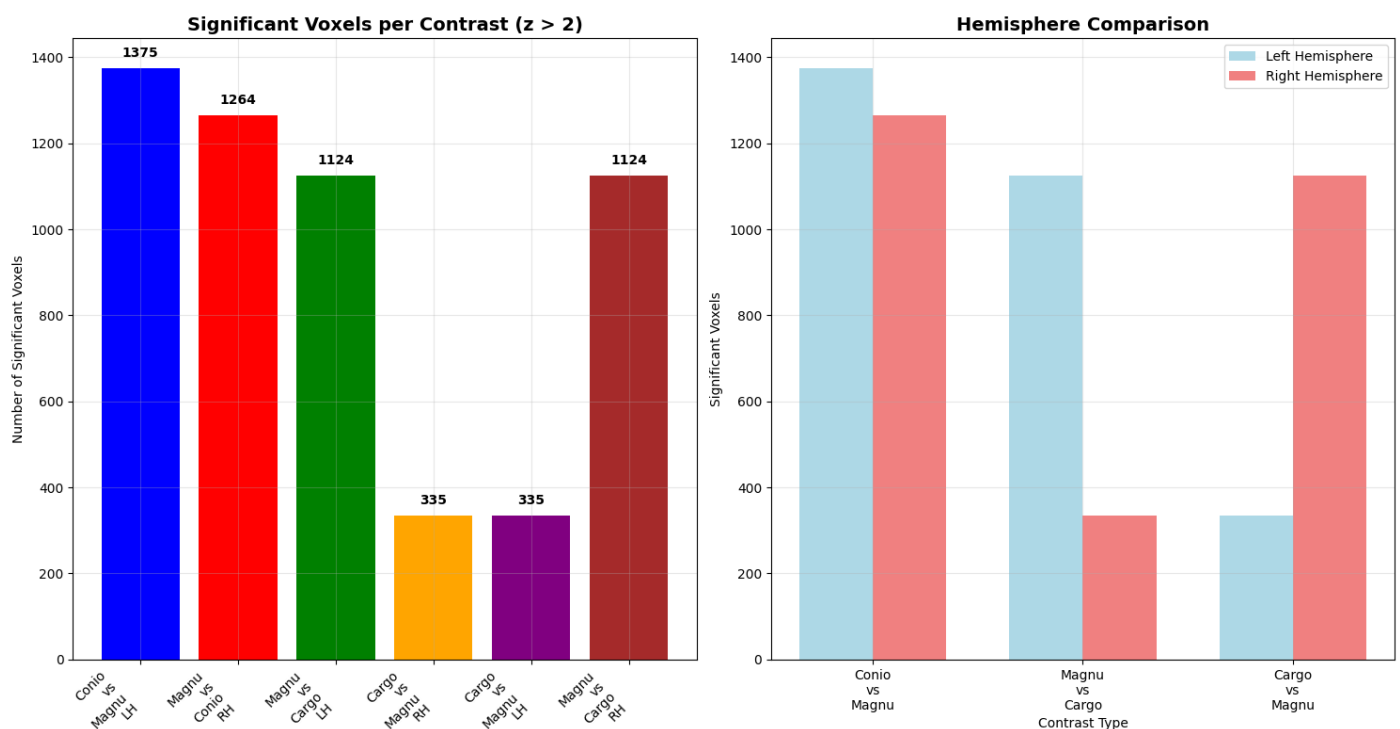


Figure 1 Figure 1 - Experimental design - Event matrix